

NeuroAnalyzer report: MUA Analysis

NeuroAnalyzer v0.3.0 · MATLAB 2026a · 2026-09-25T05:27:51+00:00 · window image: exportapp

MUA Analysis

Detect and sort spikes from multi-unit activity, then check quality and firing rate

7 Help

Channel: Ch 4

☐ Segment by alignment metrics

Segment:

40 (max 1000)

3 Spike sorting

MAD, k = 4, negative polarity, PCA + K-means, auto-merge ≥ 0.80

Configure...

Run

4 Clusters

Cluster 1: 356 spikes
Cluster 2: 177 spikes
Cluster 4: 62 spikes

Select all

Clear

Undo

Merge selected

Split selected

Ch 4: 2 units, 3 accepted, 0 rejected (SNR < 2 or > 99.99% ISI)
Unit 1: 0 noise spikes

5 Export

Save results...

Save session...

Open session...

Report (PDF)...

Methods report...

Signal & spikes

Waveforms

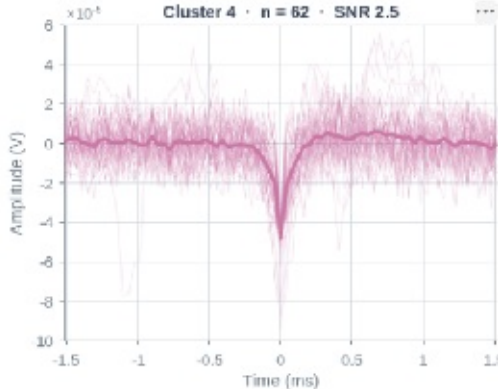
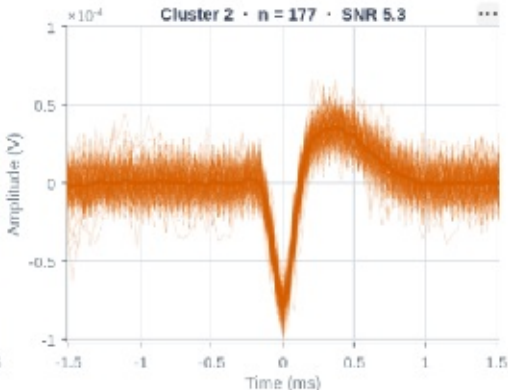
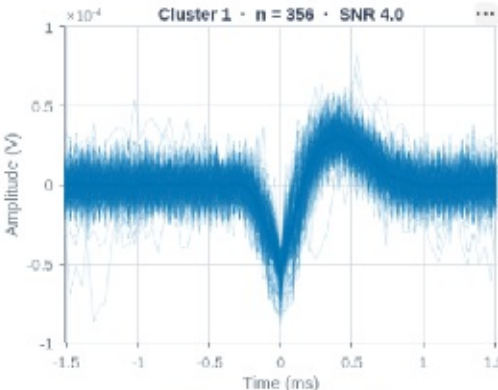
Clusters (feature space)

Spikes rate

Raster & PSTH

Correlograms

Quality



✓ Session saved to MUAAnalysisApp\session.mat (1 input file(s) with 125, 3 settings and results).

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Window: MUA Analysis (MUAAnalysisApp)
NeuroAnalyzer v0.3.0
MATLAB 26.1.0.3346908 (R2026a) Update 5
OS: Linux 6.17.0-1022-azure (GLNXA64)
Session created: 2026-09-25T05:27:51+00:00
Report created: 2026-09-25T05:27:51+00:00

NOTES
MUAAnalysisApp walkthrough: demo data

INPUTS (1)
1. MUA file: demo_mua.mat
path: /tmp/NeuroAnalyzerDemo/demo_mua.mat
14,434,714 bytes, modified 2026-09-25T05:22:18
MD5 0e6d18586e3ee89e49a38efc6ed50fab

SETTINGS
channelIndex = 2
channel = 4
segmentation.on = false
segmentation.params = []
segmentation.segments = []
segmentation.segment = 0
sortParams.detectMethod = MAD
sortParams.clusterMethod = K-means
sortParams.threshold = 4
sortParams.refractoryMs = 1
sortParams.alignWinMs = 1.5
sortParams.polarity = negative
sortParams.filter = 0
sortParams.bpLow = 300
sortParams.bpHigh = 3000
sortParams.featureMethod = PCA
sortParams.numComponents = 3
sortParams.normalize = 1
sortParams.minSpikesPerCluster = 20
sortParams.randomSeed = 0
sortParams.enableDriftCorrection = 0
sortParams.driftMethod = Time Binning
sortParams.driftBinwidth = 30
sortParams.enableGridSearchCheck = 0
sortParams.dbscanEpsilon = NaN
sortParams.autoMerge = 1
sortParams.mergeThreshold = 0.95
sortParams.mergeMinAmpRatio = 0.85
raster.from = -0.1
raster.to = 0.3
raster.binMs = 5
correlogram.maxLagMs = 50
correlogram.binMs = 1
rate.binS = 1
rate.relative = false
selectedClusters = [1 2 4]
tab = Waveforms

KEY RESULTS
MUA: 3 channel(s) at 24414.0625 Hz, analysed Ch 4
Sorted 595 spikes on Ch 4: 3 unit(s), 0 rejected
unit 1: 356 spikes, SNR 3.98, ISI < refractory 0.00%
... 4 more line(s): see the session file